# MUHAMMAD HAFIY IMRAN

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## PROFESSIONAL SUMMARY

High-achieving Intelligent Systems Engineering student (**CGPA 3.82**, 3-time Dean's List) with practical experience in backend automation, machine learning modelling, and heuristic optimisation. Demonstrated capability in building end-to-end data pipelines and operational automation tools during an enterprise internship at TM Technology Services. Seeking a data analytics or AI internship for the March–June 2027 period to apply multi-objective optimisation, predictive analytics, and algorithmic engineering to solve real-world industry problems.

## CORE SKILLS & COMPETENCIES

- **Technical & Analytical Skills:** Data Analytics, Machine Learning, Optimisations, Time Series Analysis & Forecasting, Database Management, Fuzzy Inference Systems, IoT Integration.
- **Programming & Tools:** Python, SQL, JavaScript, MySQL, Flask, Node.js, Arduino IDE, Git/GitHub, Tailwind CSS.
- **Languages:** Bahasa Melayu (Native), English (Full Professional Working Fluency).

## TECHNICAL PROJECTS & RESEARCH

**Multi-Portfolio Optimisation for Shariah-Compliant Instruments (Final Year Project)** | UiTM Shah Alam
*September 2025 – Present*

- Spearheaded the design and implementation of a multi-objective portfolio optimisation engine using the **NSGA-II (Non-dominated Sorting Genetic Algorithm II)** in Python.
- Modelled historical pricing and quantitative financial risk metrics for Malaysian Shariah-compliant equities to generate optimal Pareto-efficient asset allocations.
- Automated data retrieval and data cleaning pipelines to evaluate risk-adjusted return profiles across various market conditions.

**Automated Monkey Detection & IoT Alert System** | Academic Project
*UiTM Shah Alam*

- Architected an automated edge-detection pipeline integrating an Arduino-controlled motorised CCTV system with motion sensors to track localised wildlife movement.
- Programmed a computer vision inference workflow using Flask to classify captured images and trigger instant alert payloads via the Telegram Bot API.
- Minimised false-positive detection rates through refined sensor alignment and image preprocessing scripts.

**Hierarchical Fuzzy Inference System for Home Energy Efficiency** | Academic Project
*UiTM Shah Alam*

- Developed an intelligent diagnostic web application using Node.js and Tailwind CSS powered by a Hierarchical Fuzzy Inference System (FIS).

- Translated non-linear environmental and appliance consumption variables into clear linguistic rules to deliver actionable household energy reduction recommendations.

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# PROFESSIONAL & PRACTICAL EXPERIENCE

**SOC Intern** | TM Technology Services Sdn Bhd — Kuala Lumpur
*March 2024 – August 2024*

- Embedded within the SOC Premium Account division under Group Network Technology to streamline network incident tracking and service escalation workflows.
- Developed a centralised web tool utilising **Flask and MySQL** that automated ticket escalation and asset verification for tier-level network engineers.
- Accelerated incident response turnaround times by eliminating manual cross-referencing between ticketing records and internal configuration databases.

**Front Desk & Operational Associate** | Mail Boxes Etc. (MBE) — Seri Kembangan, Selangor
*Short-Term Practical Experience*

- Coordinated inbound and outbound courier logistics, intake verifications, and Shopee parcel collection point workflows, handling high daily transaction volumes.
- Maintained 100% accuracy in physical parcel sorting and tracking record synchronisation.

**Service & Cashier Crew** | F&B Retail — Putrajaya
*Short-Term Practical Experience*

- Handled point-of-sale (POS) terminal transactions, inventory reconciliation, and front-line customer assistance during peak operating hours at RnR MEX Putrajaya.

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# EDUCATION & QUALIFICATIONS

**Bachelor of Information Systems (Hons.) Intelligent Systems Engineering** | Universiti Teknologi MARA (UiTM), Shah Alam
*October 2025 – Expected June2027*

- **Academic Standing:** CGPA 3.82 (Dean's List Award recipient for 3 semesters).
- **Relevant Coursework:** Machine Learning, Data Mining, Time Series Analysis and Forecasting, Business Data Analytics, Programming for Data Science, Optimisation Algorithms & Applications, Database Management Systems.

**Diploma in Computer Science** | Universiti Teknologi MARA (UiTM), Kampus Arau, Perlis
*October 2022 – February 2025*

- **Academic Standing:** CGPA 3.88 (Dean's List Award recipient for 4 semesters).
- Graduated with a strong foundation in object-orientated programming, Data Communication & Networking, and Information Systems Development.

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# CERTIFICATIONS & ADDITIONAL DETAILS

- **Driving Licenses:** Valid Malaysian Driving License (Class D and B2).
- **Internship Availability:** March 2027 – June 2027 (Full-time availability).